## SEND5232: Software Architecture and Design Project

## Bereket Kiros – ugr/189518/16 – Section 2

## PHASE 2 — Architecturally Significant Requirements (ASRs)

## 📚 Lihq’et Bookstore System

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## 1. Introduction

The Architecturally Significant Requirements (ASRs) for the Lihq’et Bookstore System define the key functional and non-functional requirements that directly influence the system’s architectural design. These requirements are critical in shaping decisions related to system structure, technology selection, modularity, and scalability.

The system is a full-stack web-based bookstore management application designed using a 3-tier layered architecture combined with MVC and Service Layer patterns. The ASRs are categorized based on quality attributes such as performance, security, scalability, maintainability, and usability.

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### 2. ASR Classification Overview

The ASRs are grouped into the following quality attributes:

- Performance
- Security
- Scalability
- Maintainability
- Usability

Each requirement includes:

- ID
- Description
- Quality Attribute
- Priority Level (High / Medium / Low)

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## 3. ASRs List

### 3.1 Performance Requirements

**ASR-P1**

- **Description:** The system shall retrieve and display book records within a short response time when requested by the user.
- **Quality Attribute:** Performance
- **Priority:** High

**ASR-P2**

- **Description:** The system shall support asynchronous communication between frontend and backend using REST API without blocking user interaction.
- **Quality Attribute:** Performance
- **Priority:** High

**ASR-P3**

- **Description:** The system shall minimize redundant database queries by efficiently handling CRUD operations through service layer abstraction.
- **Quality Attribute:** Performance
- **Priority:** Medium

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### 3.2 Security Requirements

**ASR-S1**

- **Description:** The system shall ensure that only valid and properly structured input data is processed by the backend.
- **Quality Attribute:** Security
- **Priority:** High

**ASR-S2**

- **Description:** The system shall sanitize all user inputs before executing database operations to prevent injection attacks.
- **Quality Attribute:** Security

- **Priority:** High

**ASR-S3**

- **Description:** The system shall restrict direct access to the database layer, allowing interaction only through backend services.
- **Quality Attribute:** Security
- **Priority:** High

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**3.3 Scalability Requirements**

**ASR-SC1**

- **Description:** The system architecture shall support future expansion to include additional modules such as user authentication and online ordering.
- **Quality Attribute:** Scalability
- **Priority:** High

**ASR-SC2**

- **Description:** The backend shall be designed in a modular structure (Controller-Service-Model) to allow independent scaling of components.
- **Quality Attribute:** Scalability
- **Priority:** Medium

**ASR-SC3**

- **Description:** The system shall support increased number of book records without significant degradation in performance.
- **Quality Attribute:** Scalability
- **Priority:** Medium

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### 3.4 Maintainability Requirements

**ASR-M1**

- **Description:** The system shall follow separation of concerns using MVC and Service Layer architecture to improve code maintainability.
- **Quality Attribute:** Maintainability
- **Priority:** High

**ASR-M2**

- **Description:** The system shall implement reusable service functions to avoid duplication of business logic (DRY principle).
- **Quality Attribute:** Maintainability
- **Priority:** High

**ASR-M3**

- **Description:** The system shall maintain a clean and modular folder structure for frontend and backend components.
- **Quality Attribute:** Maintainability
- **Priority:** Medium

### 3.5 Usability Requirements

**ASR-U1**

- **Description:** The system shall provide a simple and intuitive user interface for managing books without requiring technical knowledge.
- **Quality Attribute:** Usability
- **Priority:** High

**ASR-U2**

- **Description:** The system shall provide real-time feedback for user actions such as adding, updating, or deleting books.
- **Quality Attribute:** Usability
- **Priority:** High

**ASR-U3**

- **Description:** The system shall include a visually structured table layout for easy reading and navigation of book records.
- **Quality Attribute:** Usability
- **Priority:** Medium

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## 4. Summary of ASRs

The ASRs defined above directly influence the architectural decisions of the Lihq'et Bookstore System. High-priority requirements such as security, performance, and maintainability have driven the adoption of a layered architecture with MVC and Service Layer patterns. These requirements ensure that the system remains efficient, secure, and extensible for future enhancements.

# Summary Table

| Category | High | Medium | Low |
|---|---|---|---|
| Performance | 2 | 1 | 0 |
| Security | 3 | 0 | 0 |
| Scalability | 1 | 2 | 0 |
| Maintainability | 2 | 1 | 0 |
| Usability | 2 | 1 | 0 |